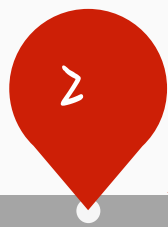
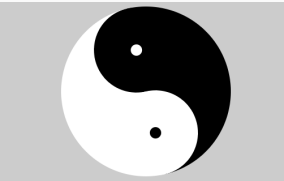


数据库

《数据库原理及应用》



关系数据库复习



关系名

关系模式

关系

元组

行数为基数

学生基本信息登记表

列个数为元

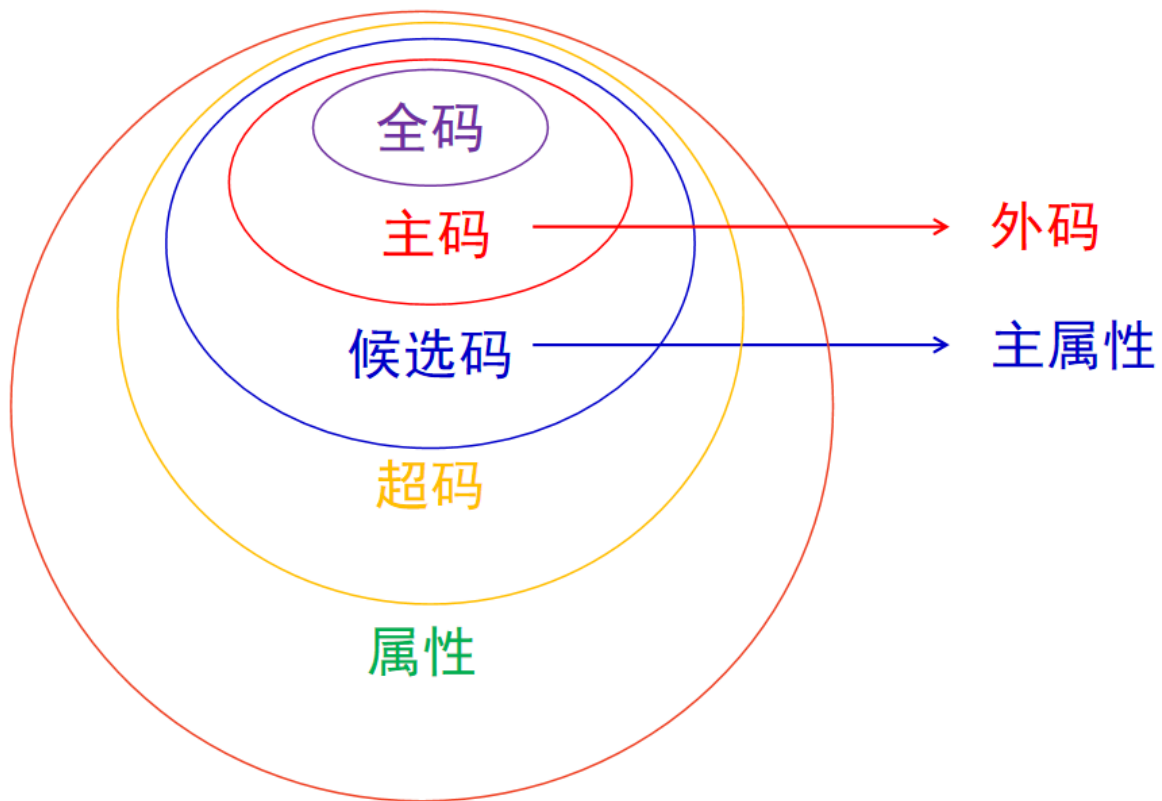
列取值为域

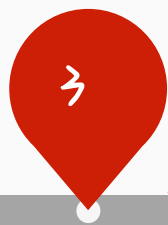
属性

学号	姓名	性别	年龄
字符型	字符型	布尔型	短整型
802	李四	男	20
803	王五	男	20
804	赵六	女	20
805	钱七	男	19

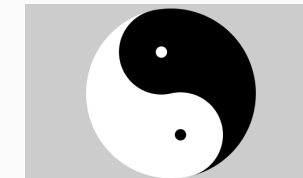
一个分量

码：那些属性组能确定一行内容？





关系数据库复习



数据结构

关系名

关系模式

关系

元组

行数为基数

学生基本信息登记表

列个数为元

列取值为域

属性

学号	姓名	性别	年龄
字符型	字符型	布尔型	短整型
802	李四	男	20
803	王五	男	20
804	赵六	女	20
805	钱七	男	19

一个分量

域

域

乘

笛卡尔积

子集

关系

域是一组具有相同数据类型的值的集合。

$$D_1 \times D_2 \times \cdots \times D_n = \{ (d_1, d_2, \cdots, d_n) \mid d_i \in D_i \}$$

记做: $R(U, D, DOM, F)$

简记: $R(U)$ 或 $R(A_1, A_2, \cdots, A_n)$

码: 那些属性组能确定一行内容?

全码

主码

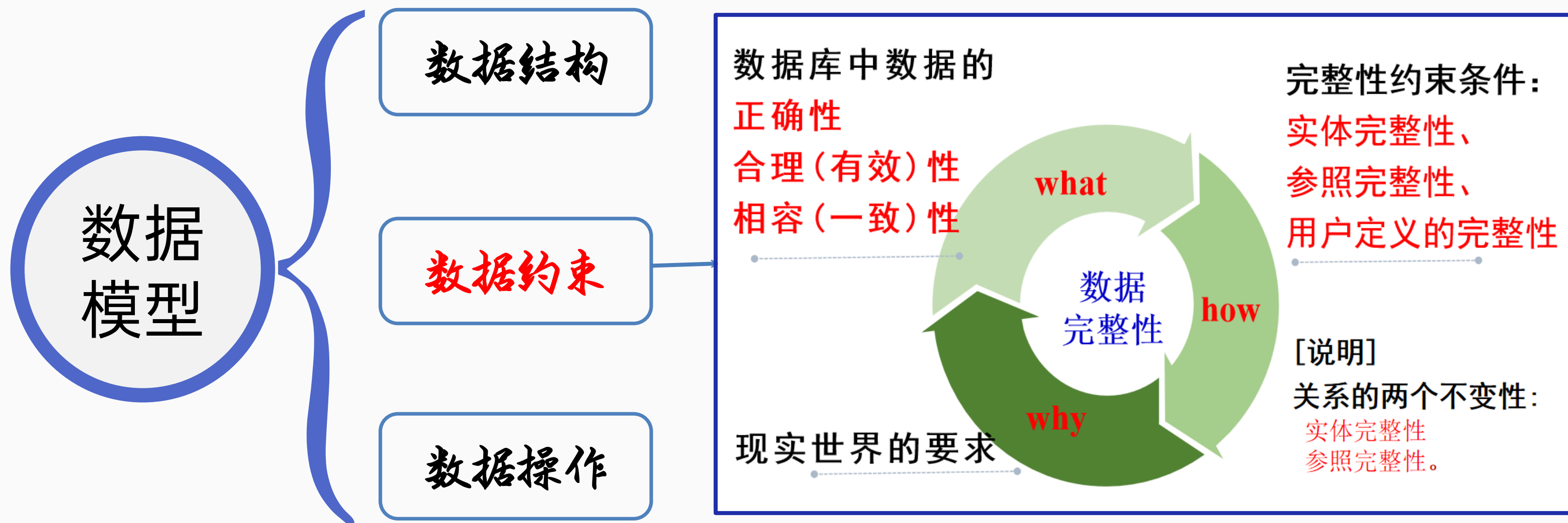
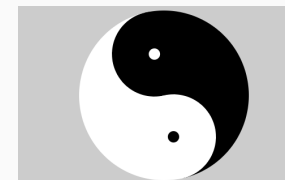
候选码

超码

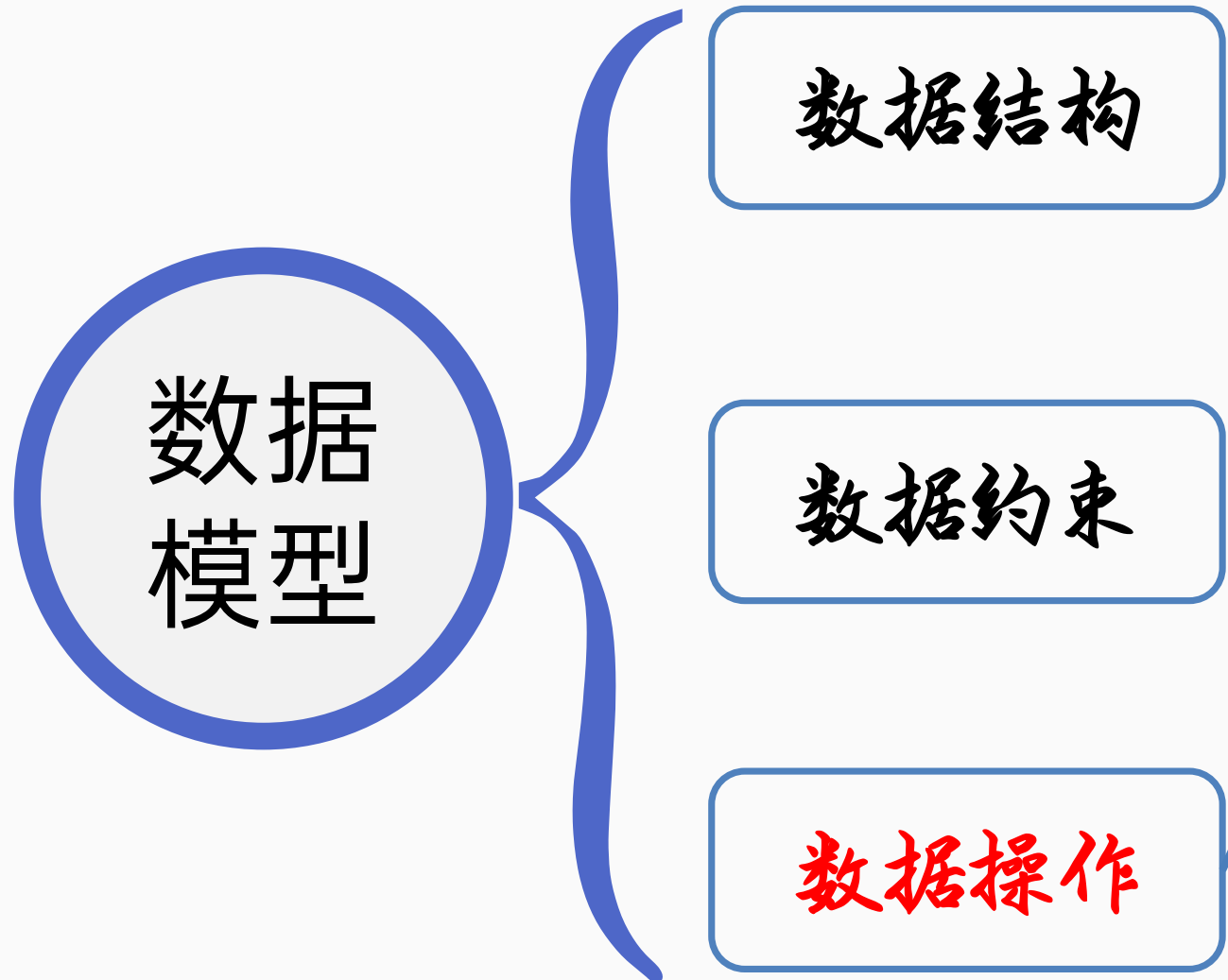
属性

外码

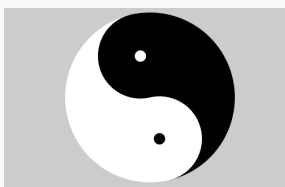
主属性



关系代数



运算符		含义	运算符		含义
集合运算符	U	并	比较运算符	>	大于
	-	差		≥	大于等于
	∩	交		<	小于
	×	笛卡尔积		≤	小于等于
专门的关系运算符	σ	选择		=	等于
	π	投影		<>	不等于
	⋈	连接	逻辑运算符	¬	非
	÷	除		∧	与
				∨	或



目

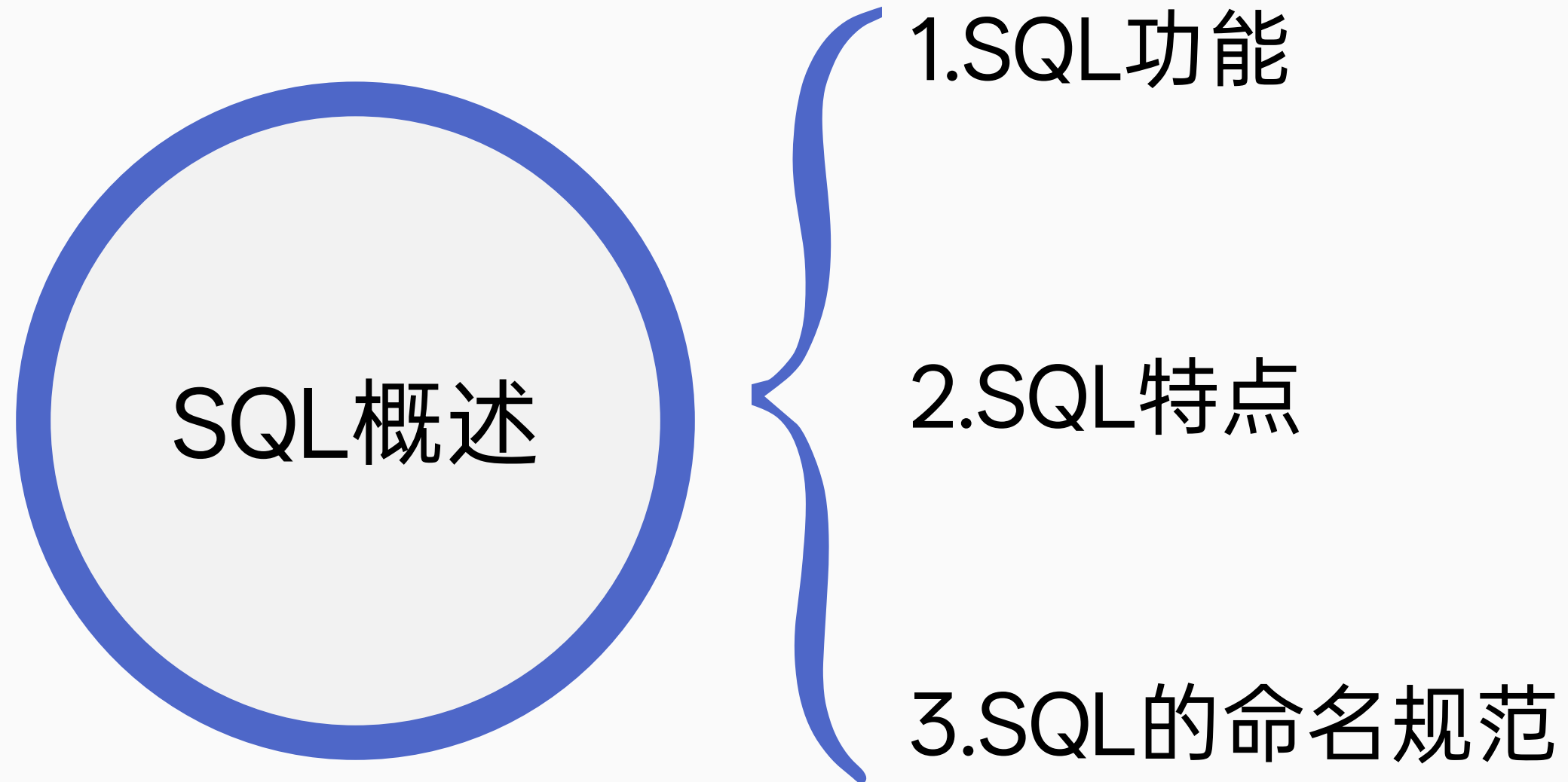
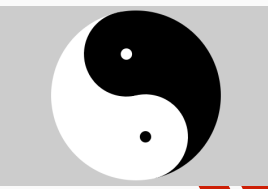
录

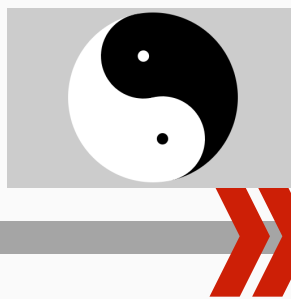
01 MySQL数据库

02 MySQL数据库的安装与配置

03 MySQL服务

04 MySQL客户端工具

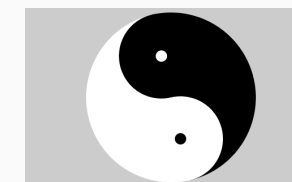




结构化查询语言 (Structured Query Language , SQL)
是关系数据库的标准语言，
是通用的、功能极强的关系数据库语言。

SQL 主要功能	动词
数据定义语言DDL (Data Definition Language)	CREATE, DROP, ALTER
数据操纵语言DML (Data Manipulation Language)	SELECT, INSERT, UPDATE, DELETE
数据控制语言DCL (Data Control Language)	GRANT, REVOKE

要求：增删改查，CRUD即Create、Read、Update、Delete。



1、关系数据库标准语言

SQL86- > 89- > 92 (SQL2)- > 99 (SQL3)- > 2003- >

业界主要实现了SQL92标准，标准外语法细节不相同。

2、综合统一，功能强大

3、高度非过程化,类似自然语言

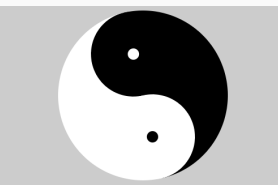
4、面向集合的操作方式

5、语言简洁，易学易用

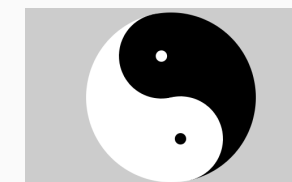
参考文档

《Java开发手册 | 第五章 MySQL 数据库》
关键字全部
使用大写字母
自定义名称
使用小写英文+
下划线隔开

对象	命名规则
数据库名(Database)	db_系统名
表 (Table)	t_英文名词单数
字段(Column)	不用表前缀, 英文单词
主键(Primary key)	pk_表名(复合主键用字段)
外键(Foreign key)	fk_从表名_主表名
Check约束(Check Constraint)	ck_表名_字段名
Default约束(Default Constraint)	df_表名_字段名
唯一约束(Unique key)	uk_表名_字段名
索引(Index)	idx_字段名
视图 (View)	view_表名1_表名2
触发器(Trigger)	trg_表名_后缀(I U、D)
函数 (function)	fun_
存储过程 (Stored procedure)	prd_



CREATE	修改表或视图中的数据
ALTER	从表或视图中删除数据
DROP	用于授予权限
SELECT	用于收回权限
INSERT	创建数据库或数据库对象
UPDATE	对数据库或数据库对象进行修改
DELETE	删除数据库或数据库对象
GRANT	从表或视图中检索数据
REVOKE	将数据插入到表或视图中



下列属于MySQL注释语法的是(A B C)

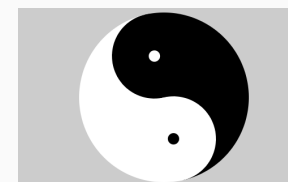
A. #我是单行注释

B. -- 单行注释 “--” 之后必须要有一个空格

C. /*

我是
多行
注释

*/



◆ 课程使用操作系统要求：

Windows11 或 Windows10

◆ 本课程采用的数据库为：

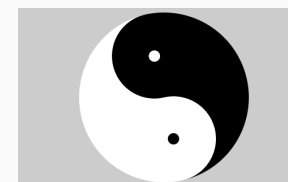
社区版 MySQL8.0.X

◆ 辅助软件：

Navicat16或Navicat17

◆ 教学数据库：

db_sc（学生表、课程表、选课表）

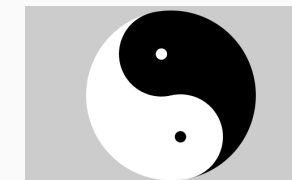


注意事项

- ◆ 对操作系统**不建议**：使用中文用户名
- ◆ 对安装目录**不建议**：有中文和空格

建议：D:\mysql8 和 D:\navicat16

- ◆ MySQL字符集建议：选择`utf8mb4`
- ◆ MySQL的密码建议：使用Root
- ◆ Navicat使用建议：即时刷新数据库
- ◆ 教学数据库的建议：先删除外键约束



社区版 MySQL8.0 的安装演示：

MySQL Installer

Adding Community

Choosing a Setup Type

Download

Installation

Installation Complete

Choosing a Setup Type

Please select the Setup Type that suits your use case.

- ☐ **Developer Default**
Installs all products needed for MySQL development purposes.
- ☒ **Server only**
Installs only the MySQL Server product.
- ☐ **Client only**
Installs only the MySQL Client products, without a server.
- ☐ **Full**
Installs all included MySQL products and features.
- ☐ **Custom**
Manually select the products that should be installed on the system.

Setup Type Description

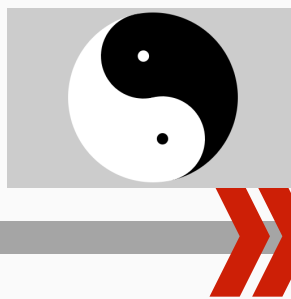
Installs only the MySQL Server. This type should be used where you want to deploy a MySQL Server, but will not be developing MySQL applications.

只安装Mysql

Next >

Cancel

CSDN@wenny小姐姐



MySQL Installer

MySQL. Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Check Requirements

The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.

	For Product	Requirement	Status
☐	MySQL Server 8.0.32	Microsoft Visual C++ 2019 Redistrib...	

Requirement Details

MySQL Installer is trying to resolve this requirement automatically.
There is nothing you need to do.

Requirement:

Microsoft Visual C++ 2019 Redistributable Package (x64) is not installed. Latest binary compatible version will be installed if agreed

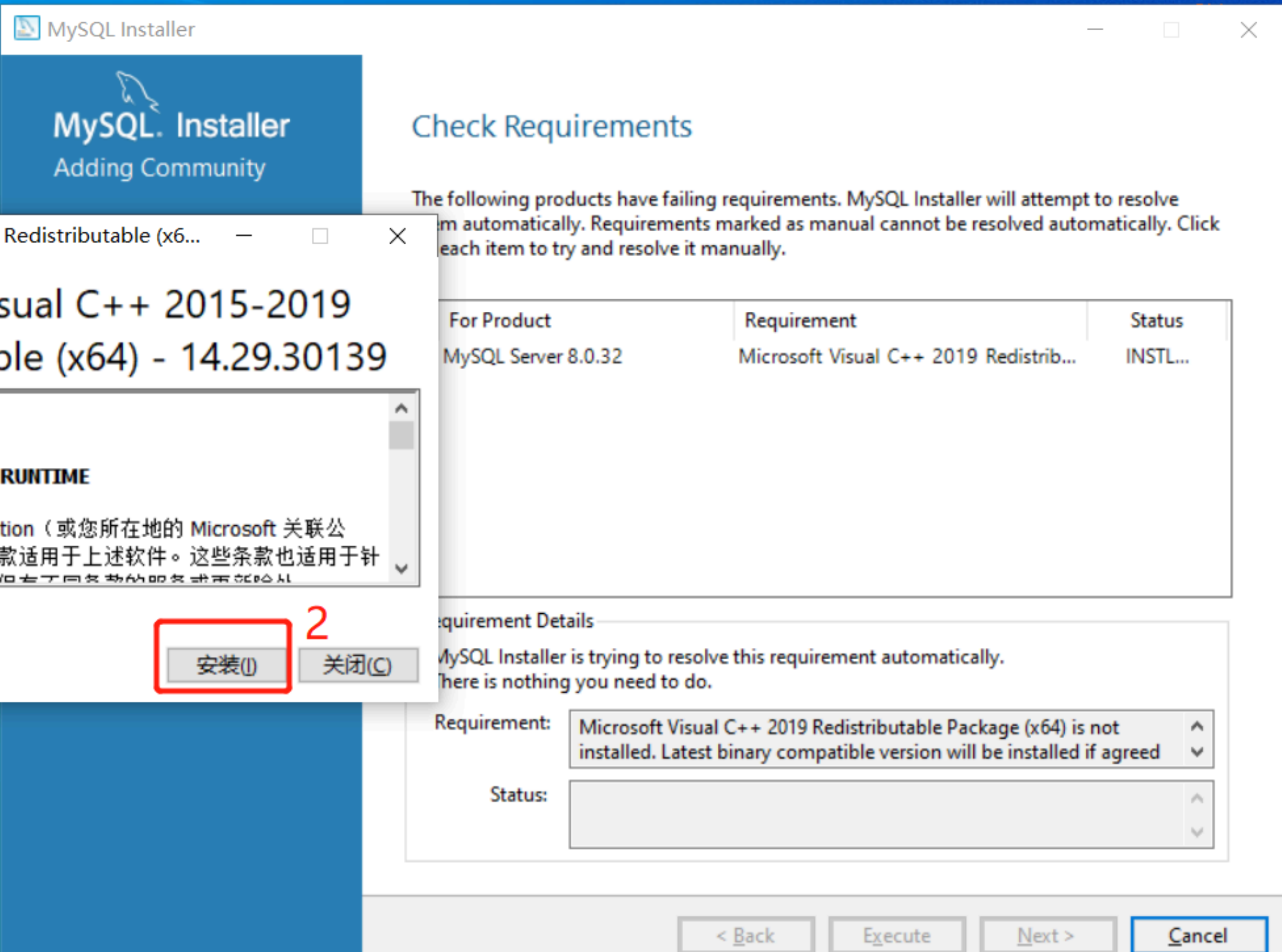
Status:

< Back

Execute

Next >

Cancel



MySQL. Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Check Requirements

The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.

For Product	Requirement	Status
✓ MySQL Server 8.0.32	Microsoft Visual C++ 2019 Redistrib...	INSTL DONE

Requirement Details

MySQL Installer is trying to resolve this requirement automatically.
There is nothing you need to do.

Requirement: Microsoft Visual C++ 2019 Redistributable Package (x64) is not installed. Latest binary compatible version will be installed if agreed

Status:

< Back

Next >

Cancel

CSDN@wenyiny小姐姐

MySQL Installer

MySQL. Installer
Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Installation

The following products will be installed.

Product	Status	Progress	Notes
 MySQL Server 8.0.32	Ready to Install		

Click [Execute] to install the following packages.

< Back

Execute

Cancel

MySQL Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Installation

The following products will be installed.

Product	Status	Progress	Notes
 MySQL Server 8.0.32	Complete		

Show Details >

< Back

Next >

Cancel

CSDN @wenny小姐

MySQL Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Product Configuration

We'll now walk through a configuration wizard for each of the following products.

You can cancel at any point if you wish to leave this wizard without configuring all the products.

Product	Status
MySQL Server 8.0.32	Ready to configure

Next >

Cancel

MySQL. Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Type and Networking

Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type: Development Computer

Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP

Port: 3306

X Protocol Port: 33060

☒ Open Windows Firewall ports for network access

☐ Named Pipe

Pipe Name: MYSQL

☐ Shared Memory

Memory Name: MYSQL

Advanced Configuration

Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.

☐ Show Advanced and Logging Options

Next >

Cancel

CSDN@wenyijiaojiao

MySQL Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Authentication Method

☐ Use Strong Password Encryption for Authentication (RECOMMENDED)

MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.



Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).

Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.

☒ Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)

Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:

- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.

Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.

< Back

Next >

Cancel

CSDN@wenyixiaozhu

MySQL Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Accounts and Roles

Root Account Password

Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

输入密码



Repeat Password:

再次输入密码

MySQL User Accounts

Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role

Add User

Edit User

Delete

< Back

Next >

Cancel

CSDN @wenny小姐姐

MySQL Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Windows Service

☒ Configure MySQL Server as a Windows Service

Windows Service Details

Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.

Windows Service Name: ☒ Start the MySQL Server at System Startup

如果出现黄色感叹号警告，表示名称重复了，可换个其他名称

Run Windows Service as ...

The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.

☒ Standard System Account

Recommended for most scenarios.

☐ Custom User

An existing user account can be selected for advanced scenarios.

< Back

Next >

Cancel

CSDN @wongy小姐姐

MySQL. Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Server File Permissions

MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:

C:\ProgramData\MySQL\MySQL Server 8.0\Data

Do you want MySQL Installer to update the server file permissions for you?

- ☒ Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
- ☐ Yes, but let me review and configure the level of access.
- ☐ No, I will manage the permissions after the server configuration.

< Back

Next >

Cancel

GSDN@wenny小姐姐

MySQL Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Apply Configuration

Click [Execute] to apply the changes

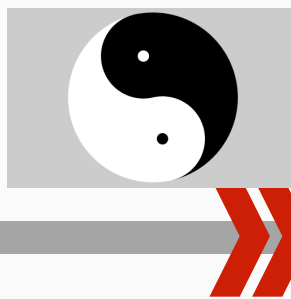
Configuration Steps Log

- ☐ Writing configuration file
- ☐ Updating Windows Firewall rules
- ☐ Adjusting Windows service
- ☐ Initializing database (may take a long time)
- ☐ Updating permissions for the data folder and related server files
- ☐ Starting the server
- ☐ Applying security settings
- ☐ Updating the Start menu link

< Back

Execute

Cancel



MySQL Installer

MySQL Server 8.0.34

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Apply Configuration

The configuration operation has completed.

Configuration Steps

Log

☒ Writing configuration file

☒ Updating Windows Firewall rules

☒ Adjusting Windows service

☒ Initializing database (may take a long time)

☒ Updating permissions for the data folder and related server files

☒ Starting the server

☐ Applying security settings

☐ Updating the Start menu link

The configuration for MySQL Server 8.0.34 has failed.

You can find more information about the failures in the 'Log' tab.

< Back

Execute

Cancel

MySQL Installer

MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Apply Configuration

The configuration operation has completed.

Configuration Steps

Log

- ✓ Writing configuration file
- ✓ Updating Windows Firewall rules
- ✓ Adjusting Windows service
- ✓ Initializing database (may take a long time)
- ✓ Updating permissions for the data folder and related server files
- ✓ Starting the server
- ✓ Applying security settings
- ✓ Updating the Start menu link

The configuration for MySQL Server 8.0.32 was successful.
Click Finish to continue.

Finish

MySQL Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Product Configuration

We'll now walk through a configuration wizard for each of the following products.

You can cancel at any point if you wish to leave this wizard without configuring all the products.

Product	Status
MySQL Server 8.0.32	Configuration complete.

Next >

Cancel

MySQL. Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

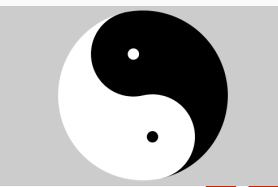
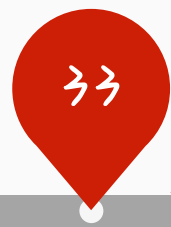
Installation Complete

Installation Complete

The installation procedure has been completed.

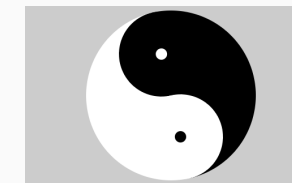
Copy Log to Clipboard

Finish

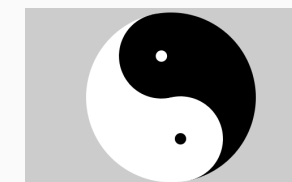


MySQL有四个自带的数据库

数据库	描述
mysql	存储MySQL服务器正常运行所需要的各种信息 (时区、主从、用户、权限等)
information_schema	提供了访问数据库元数据的各种表和视图， 包含数据库、表、字段类型及访问权限等
performance_schema	提供了MySQL服务器运行时状态底层监控功能， 主要用于收集数据库服务器性能参数
sys	包含了一系列方便 DBA 和开发人员利用 <i>performance_schema</i> 性能数据库进行性能调优和诊断的视图

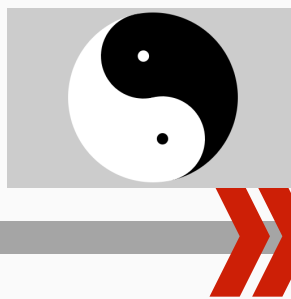


Navicat17 的安装演示



Navicat17 的使用演示

1. 数据库的连接
2. 数据库的创建和删除
3. 表的创建（选择数据类型）、修改和删除
4. 表数据的增删改查
5. 表的导入导出（SQL文件、文本文件、表格文件等）
6. 数据库的备份还原



Student表

学号 Sno	姓名 Sname	性别 Ssex	年龄 Sage	所在系 Sdept
201215121	李勇	男	20	CS
201215122	刘晨	女	19	CS
201215123	王敏	女	18	MA
200515125	张立	男	19	IS

投影列

筛选行

分组统计?

男女
年龄

导入教学演示数据库 db_sc

student表

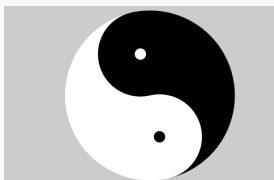
学号 <i>Sno</i>	姓名 <i>Sname</i>	性别 <i>Ssex</i>	年龄 <i>Sage</i>	所在系 <i>Sdept</i>
201215121	李勇	男	20	CS

course表

课程号 <i>Cno</i>	课程名 <i>Cname</i>	先行课 <i>Cpno</i>	学分 <i>Ccredit</i>
1	数据库	5	4

SC表

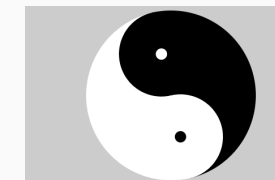
学号 <i>Sno</i>	课程号 <i>Cno</i>	成绩 <i>Grade</i>
201215121	1	92



SELECT <目标列表表达式> *FROM* <表名>

目标列表表达式：

- ① 全部列：*
- ② 列名：列名1, 列名2
- ③ 去掉重复值：Distinct 列名
- ④ 列别名：列名1 [AS] 列别名
- ⑤ 常量值：' 中国 ', 100
- ⑥ 表达式：2022 - 列名
- ⑦ 函数：YEAR(NOW())



例1、查询全体学生姓名。

```
SELECT Student.Sname  
FROM Student;
```

```
SELECT Sname  
FROM Student;
```

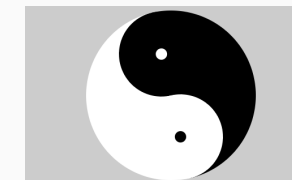
例2、查询全体学生的信息(姓名在前)。

```
SELECT Sname , Sno , Ssex, Sage, Sdept  
FROM Student;
```

例3、查询全体学生信息。

```
SELECT *  
FROM Student;
```

只在测试中使用



例4、查询全体学生年龄。

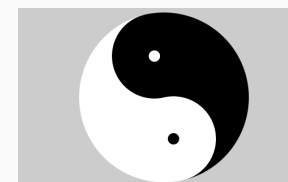
```
SELECT Sage  
FROM student;
```

例5、查询全体学生年龄(去掉重复的值)。

```
SELECT DISTINCT Sage  
FROM student;
```

例6、查询学生的性别，年龄(去掉重复行)。

```
SELECT DISTINCT Ssex ,Sage  
FROM student;
```



例7、查询学生的信息(添加‘理工学院’列)。

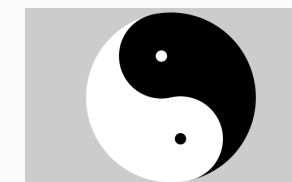
```
SELECT '理工学院', student.*  
FROM student;
```

例8、查询学生的信息(添加学校列，值为‘理工学院’)。

```
SELECT '理工学院' AS school, student.*  
FROM student;
```

例9、查询全体学生姓名(添加学校列，列名改为汉字)。

```
SELECT '理工学院' 学校, sname 姓名  
FROM student;
```



例10、查询全体学生的姓名及其出生年份。

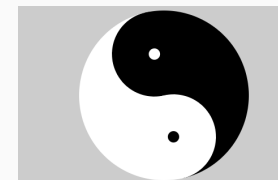
```
SELECT Sname , 2025-Sage  
FROM Student;
```

例11、查询全体学生的姓名及其出生年份（*birthday*列）。

```
SELECT Sname , 2025-Sage AS birthday  
FROM Student;
```

例12、查询全体学生的姓名及其出生年份（使用函数）。

```
SELECT Sname , YEAR(NOW ())-Sage AS  
birthday
```

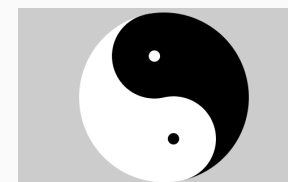


例13、查询全体学生的姓名、年龄、是否成年。

```
SELECT sname,sage,if(sage-18>=0,'Y','N') FROM  
student;
```

例14、查询全体学生的姓名、性别、颜色标识（男蓝女红）。

```
SELECT Sname , Ssex,  
CASE Ssex  
WHEN '男' THEN '蓝色'  
WHEN '女' THEN '红色'  
ELSE '其他'  
END  
FROM Student;
```



常用系统内置函数

聚合函数: `Count()`、`Sum()`、`Max()`、`Min()`、`Avg()`

数学函数: `Rand()`、`Mod(,)`、`Pow(,)`、`Sqrt()`、`Abs()`

字符串函数: `LENGTH()`、`CHAR_LENGTH`、`CONCAT()`

、

`UPPER()`、`LOWER()`、`SUBSTRING()`

日期时间函数: `NOW()`、`YEAR()`、`CURTIME()`

条件判断函数: `IF`、`CASE`

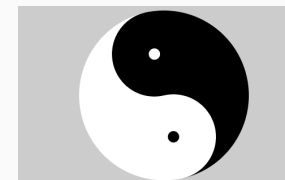
系统信息函数: `VERSION()`、`DATABASE()`、`USER()`

加密函数: `ENCODE(,)`、`DECODE(,)`、`MD5()`

格式化函数: `ASCII()`、`HEX()`、

`INET_ATON('192.168.0.1')`、

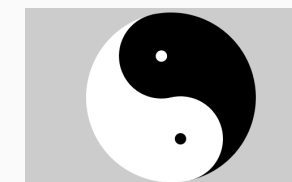
`SELECT INET_NTOA(3232235521)`



SELECT <目标列表表达式> *FROM* <表名>

目标列表表达式：

- ① 全部列：*
- ② 列名：列名1, 列名2
- ③ 去掉重复值：*Distinct* 列名
- ④ 列别名：列名1 [*AS*] 列别名
- ⑤ 常量值：' 中国 ', 100
- ⑥ 表达式：2022 - 列名
- ⑦ 函数：*YEAR(NOW())*



请在课程表上做练习，课程表结构如下：

course 表

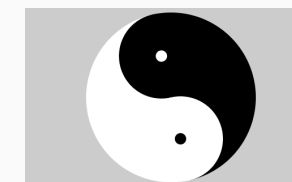
课程号 <i>Cno</i>	课程名 <i>Cname</i>	先行课 <i>Cpno</i>	学分 <i>Ccredit</i>
1	数据库	5	4

练习1、查看Course表的全部数据。

```
SELECT * FROM course ;
```

练习2、Course表中查询所有课程名，并添加开课学校列。

```
SELECT '城建大学' AS 开课学校, Cname 课程名  
FROM course ;
```



请在课程表上做练习，课程表结构如下：

course 表

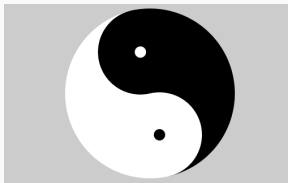
课程号 <i>Cno</i>	课程名 <i>Cname</i>	先行课 <i>Cpno</i>	学分 <i>Ccredit</i>
1	数据库	5	4

练习3、Course表中查询课程学分绩点。

```
SELECT DISTINCT Ccredit FROM course ;
```

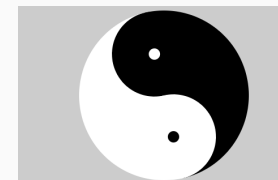
练习4、Course表中查询课程名和调整加一后的学分绩点。

```
SELECT Cname ,Ccredit+1 AS 调整后绩点  
FROM course ;
```



SELECT 列 *FROM* 表 *WHERE* 条件 ;

条件	运算符
比较	=, >, <, >=, <=, !=, <>, !>, !<;
组合	<i>AND</i> , <i>OR</i> , <i>NOT</i>
范围	<i>BETWEEN ... AND ...</i> , <i>NOT BETWEEN ... AND ..</i>
集合	<i>... IN(...)</i> , <i>... NOT IN(...)</i>
空值	<i>IS NULL</i> , <i>... IS NOT NULL</i>
模糊	<i>... LIKE ...</i> , <i>... NOT LIKE</i>
排序	<i>ORDER BY</i> 列 <i>ASC</i> , <i>ORDER BY</i> 列 <i>DESC</i>
分页	<i>LIMIT</i> (<i>SQL server</i> 类似关键字为 <i>top</i>)



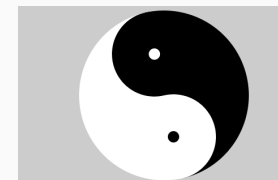
比较查询： =, >, <, >=, <=, !=, <>,

例1、查询男学生的全部信息。

```
SELECT *  
FROM student  
WHERE ssex='男'
```

例2、查询年龄在 20 岁以下的学生姓名和年龄。

```
SELECT Sname, Sage  
FROM student  
WHERE Sage < 20;
```



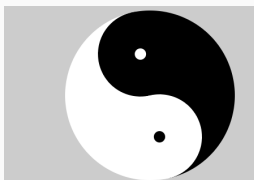
组合条件查询：AND, OR, NOT

例3、查询20岁以下的男学生姓名。

```
SELECT Sname  
FROM Student  
WHERE ssex='男' AND Sage<20;
```

例4、查询性别不是‘男’的学生信息。

```
SELECT *  
FROM Student  
WHERE NOT ssex='男';
```



范围查询: *BETWEEN...AND... NOT*

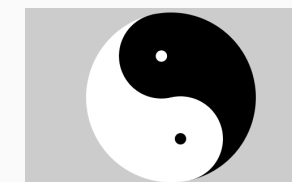
BETWEEN...AND...

例5、查询年龄在 20~23 岁之间的学生的信息。

```
SELECT *  
FROM student  
WHERE Sage BETWEEN 20 AND 23;
```

例6、查询年龄不在 20~23 岁之间的学生的信息。

```
SELECT *  
FROM student  
WHERE Sage NOT BETWEEN 20 AND 23;
```



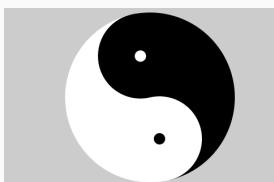
集合条件查询: ...*N*(...), ...*NOT N*(...)

例7、查询信息系 (*IS*)、数学系 (*MA*) 和计算机科学系 (*CS*) 学生的姓名和性别。

```
SELECT Sname, Ssex FROM student  
WHERE Sdept IN ('IS', 'MA', 'CS');
```

例8、查询不是信息系 (*IS*)、数学系 (*MA*) 和计算机科学系 (*CS*) 学生的姓名和性别。

```
SELECT Sname, Ssex FROM student  
WHERE Sdept NOT IN ('IS', 'MA', 'CS');
```



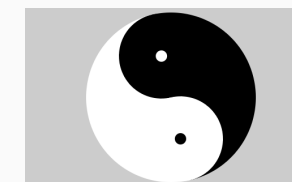
空值条件查询: *IS NULL, ... IS NOT NULL*

例9、查询缺少系别的学生信息。

```
SELECT Sno, Sname  
FROM student  
WHERE Sdept IS NULL;
```

例10、查询有系别信息的学生的学号姓名。

```
SELECT Sno, Sname  
FROM student  
WHERE Sdept IS NOT NULL;
```



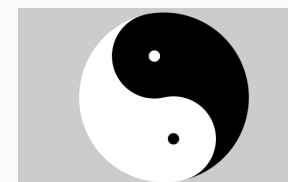
模糊查询: ...LIKE..., ...NOT LIKE...

例11、查询所有姓“刘”的学生的姓名、学号。

```
SELECT Sname, Sno  
FROM Student  
WHERE Sname LIKE '刘%'; # 百分号为多字通配符
```

例12、查询姓“刘”名字2个字组成的学生信息。

```
SELECT Sname, Sno  
FROM Student  
WHERE Sname LIKE '刘_'; # 下划线为单字通配符
```



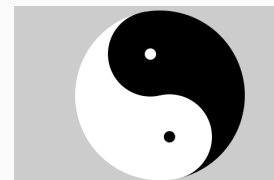
排序： *ORDER BY* 列 *ASC*； *ORDER BY* 列
DESC；

例13、查询学生姓名和年龄，结果按年龄升序排列。

```
SELECT Sname, Sage FROM student  
ORDER BY Sage ;
```

例14、查询全体学生情况，查询结果按所在系的系号升序排列，同一系中的学生按年龄降序排列。

```
SELECT * FROM student  
ORDER BY SDept, Sage DESC;
```



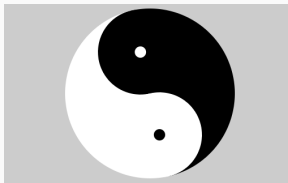
分页查询: $LMII\ m$; $LMII\ m, n$

例15、查询年龄最大的3名学生的姓名。

```
SELECT Sname  
FROM student  
ORDER BY Sage DESC  
LMII 3 ;
```

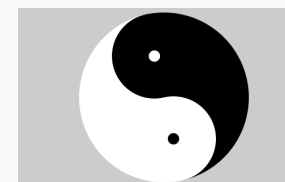
例16、查询从第3名开始4名学生的全部信息。

```
SELECT * FROM student LMII 2, 4;
```



SELECT 列 *FROM* 表 *WHERE* 条件；

条件	运算符
比较	=, >, <, >=, <=, !=, <>, !>, !<;
组合	<i>AND</i> , <i>OR</i> , <i>NOT</i>
范围	<i>BETWEEN ... AND ...</i> , <i>NOT BETWEEN ... AND ..</i>
集合	<i>... IN(...)</i> , <i>... NOT IN(...)</i>
空值	<i>IS NULL</i> , <i>... IS NOT NULL</i>
模糊	<i>... LIKE ...</i> , <i>... NOT LIKE</i>
排序	<i>ORDER BY</i> 列 <i>ASC</i> , <i>ORDER BY</i> 列 <i>DESC</i>
分页	<i>LIMIT</i> (<i>SQL server</i> 类似关键字为 <i>top</i>)



请在`course`表上做练习，课程表结构如下：

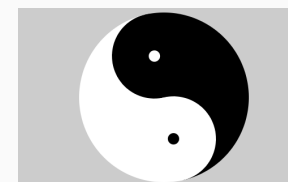
课程号 <i>Cno</i>	课程名 <i>Cname</i>	先行课 <i>Cpno</i>	学分 <i>Ccredit</i>
1	数据库	5	4

练习1、查看`Course`表的全部数据。

```
SELECT * FROM course ;
```

练习2、对`Course`表按照学分降序排列。

```
SELECT * FROM course ORDER BY Ccredit  
DESC;
```



练习3、Course表中取3条学分绩点最少的课程信息。

```
SELECT * FROM course  
ORDER BY Ccredit  
LIMIT 3 ;
```

练习4、Course表中查询没有先修课课程的全部信息。

```
SELECT * FROM course  
WHERE Cprno IS NULL ;
```

练习5、Course表中查询“数据”相关课程信息。

```
SELECT * FROM course  
WHERE Cname LIKE '%数据%';
```

学生选课表 (SC 表)

学号 <i>Sno</i>	课程号 <i>Cno</i>	成绩 <i>Grade</i>
201215121	1	92
201215121	2	85
201215121	3	88
201215122	2	90
201215122	3	80

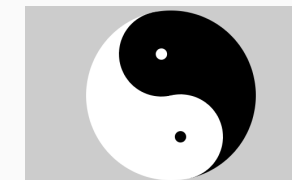
全表统计

- 总行数
- 最大值
- 最小值
- 总数值
- 平均值

不同人的平均成绩
不同课程的平均分

筛选分组

平均90分
以上的学生



分组
统计

1.全表统计 (聚合函数)

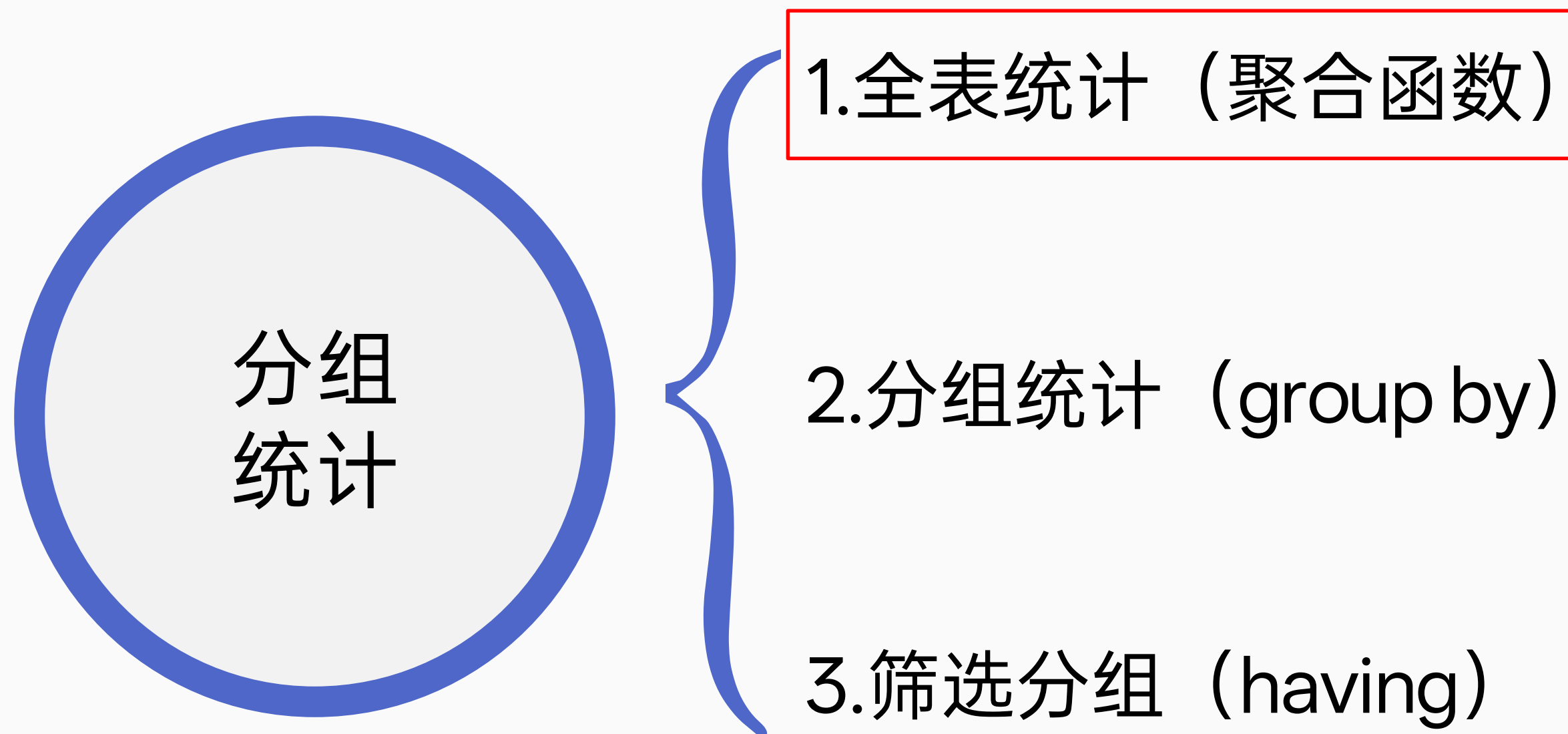
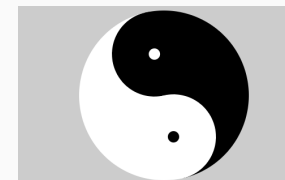
2.分组统计 (group by)

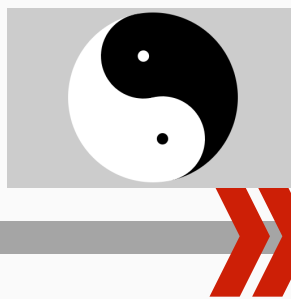
3.筛选分组 (having)

聚集函数

功能	用法
按元组计数	<code>COUNT ([DISTINCT ALL] *)</code>
按某列计数	<code>COUNT ([DISTINCT ALL] <列名>)</code>
计算总和	<code>SUM ([DISTINCT ALL] <列名>)</code>
计算平均值	<code>AVG ([DISTINCT ALL] <列名>)</code>
最大值	<code>MAX ([DISTINCT ALL] <列名>)</code>
最小值	<code>MIN ([DISTINCT ALL] <列名>)</code>

说明：如果计算时遇到空值 `NULL`，除 `COUNT(*)` 函数以外，其他函数均忽略空值。最好在计算前判断空值。

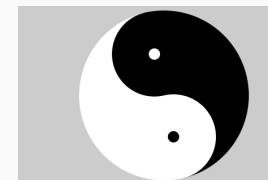




聚集函数

功能	用法
按元组计数	<code>COUNT ([DISTINCT ALL] *)</code>
按某列计数	<code>COUNT ([DISTINCT ALL] <列名>)</code>
计算总和	<code>SUM ([DISTINCT ALL] <列名>)</code>
计算平均值	<code>AVG ([DISTINCT ALL] <列名>)</code>
最大值	<code>MAX ([DISTINCT ALL] <列名>)</code>
最小值	<code>MIN ([DISTINCT ALL] <列名>)</code>

说明：如果计算时遇到空值 `NULL`，除 `COUNT(*)` 函数以外，其他函数均忽略空值。最好在计算前判断空值。



例1、查询选课总条数。

```
SELECT COUNT(*) FROM SC;
```

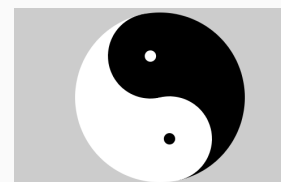
例2、查询选修了课程的学生人次。

```
SELECT COUNT(Sno) FROM SC;
```

例3、查询选修了课程的学生人数。

```
SELECT COUNT(DISTINCT Sno) FROM SC;
```

提示：统计总数使用SQL标准，`COUNT(*)`



例4、计算选修了“1”号课程的学生们的平均成绩。

```
SELECT AVG(Grade) FROM SC WHERE Cno='1';
```

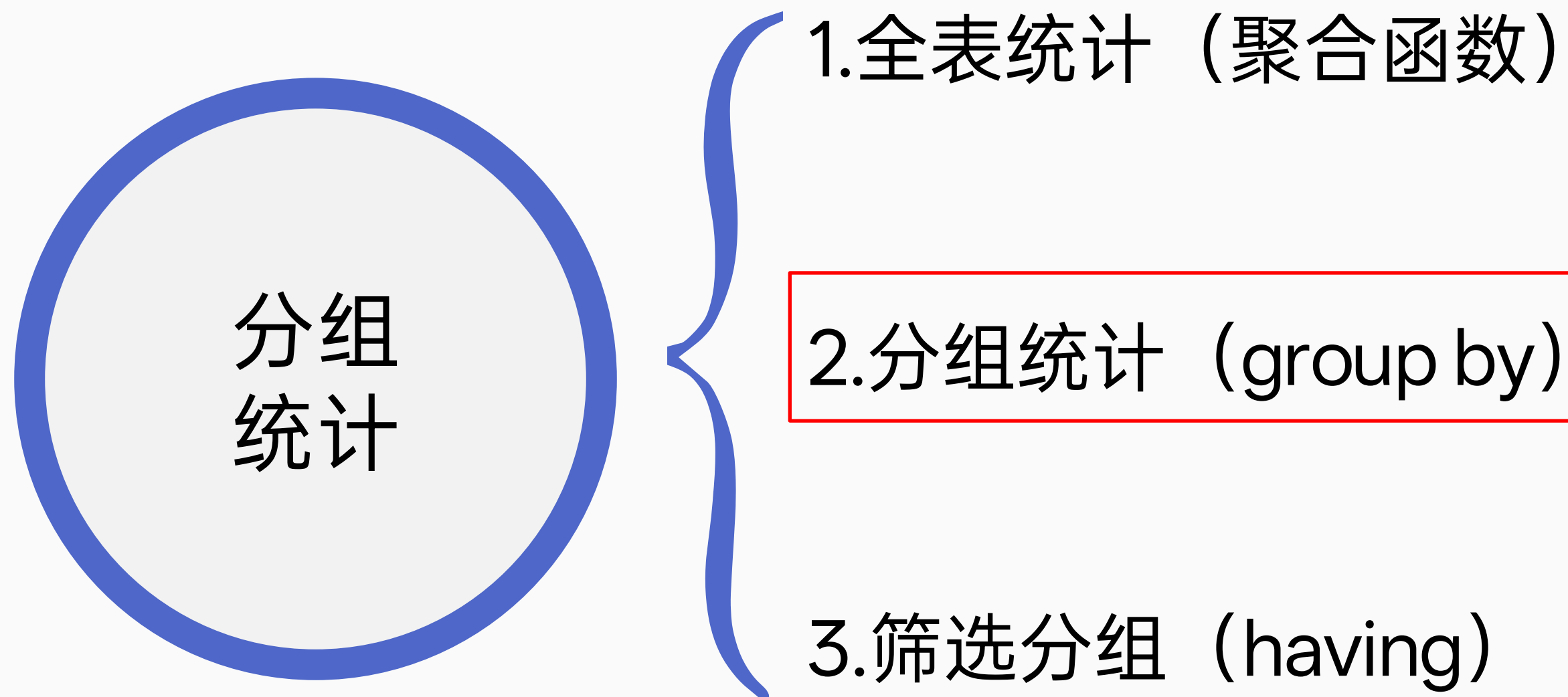
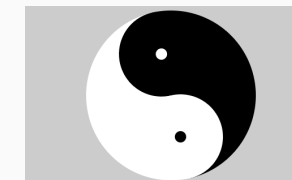
例5、查询选修“2”号课程的课程号和最高分数。

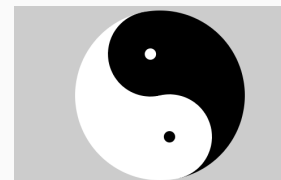
```
SELECT Cno, MAX(Grade) FROM SC WHERE
```

例6、查询选修“2”号课程的学号，课号，总分？

```
SELECT Sno, Cno, SUM(Grade) FROM SC WHERE
```

提示：使用聚合函数不应当出现统计信息内的个体信息。





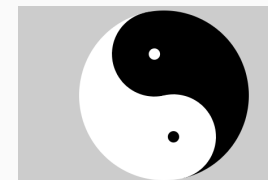
GROUP BY子句：可用于细化聚集函数的作用对象

- ◆ 按指定的一系列或多列的值分组，值相等的为一组，“相等”是指所有涉及的列的组合值相等。
- ◆ 在没有对查询结果分组时，聚集函数将作用于整个查询结果。对查询结果分组后，聚集函数将分别作用于每个组。

例7、求各门课程的课程号及相应的选课人数。

```
SELECT Cno, COUNT(Sno) FROM SC;
```

```
SELECT Cno, COUNT(Sno) FROM SC GROUP
```



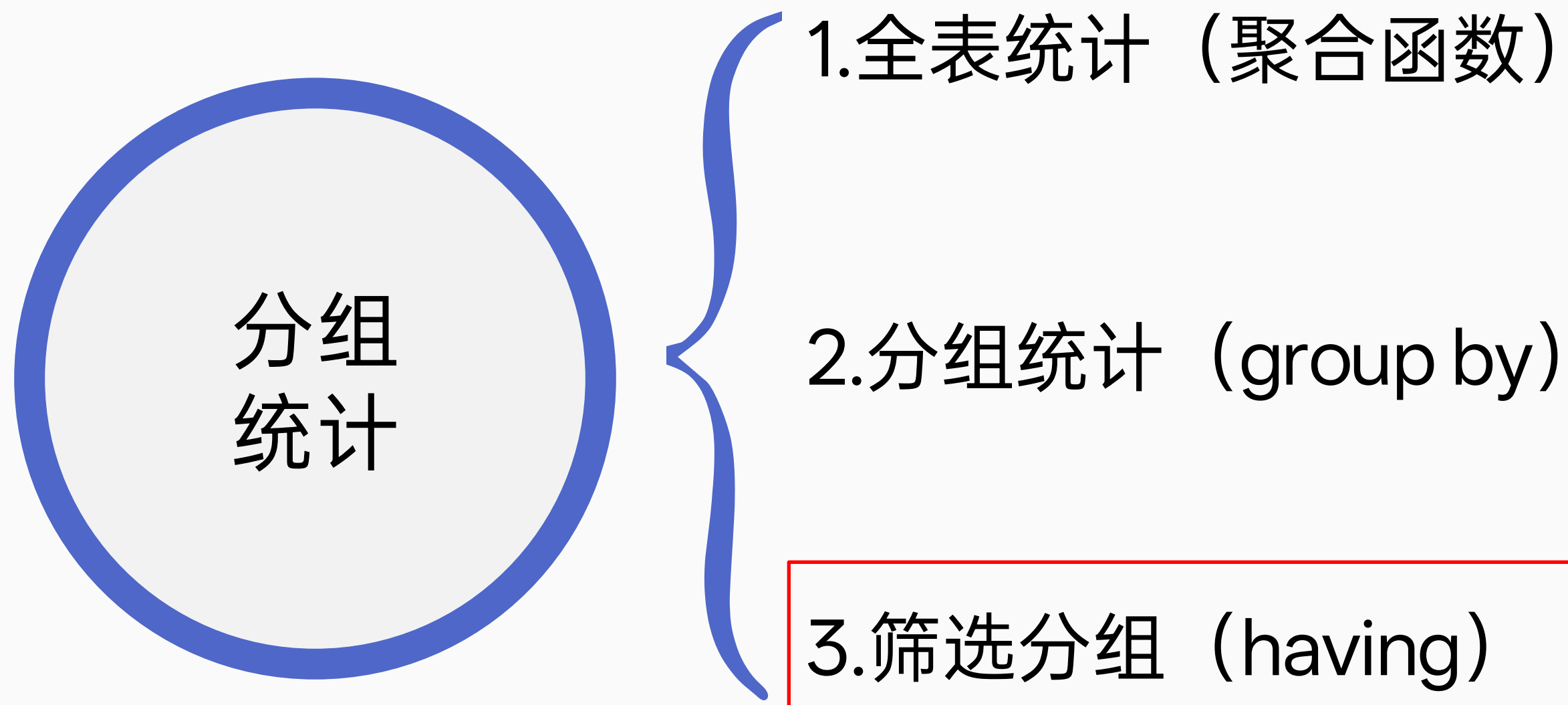
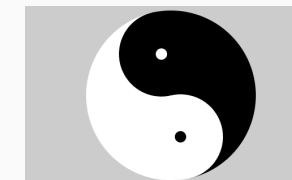
例8、求各门课程的课程号和课程平均分、最低分、最高分。

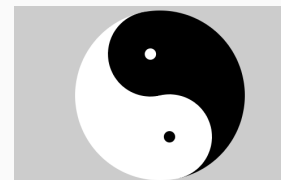
```
SELECT Cno,AVG(Grade),MAX(Grade),MAX(Grade)
FROM SC
GROUP BY Cno;
```

例9、求各学生学号及选课门数、平均分、最低分。

```
SELECT Sno,count(*),AVG(Grade),MIN(Grade)
FROM SC
GROUP BY Sno;
```

提示：GROUP BY需配合聚合函数使用，用以获取统计信息。



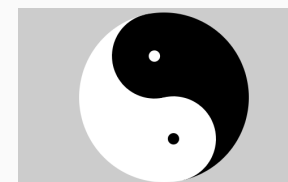


Having 子句:

当查询满足某些条件的分组数据时，无法用 *Where* 子句书写条件，因为此时的条件是需要先进行分组计算才能进行判断的，这种条件需要使用 *Having* 子句来完成。

例10、查询选修了 2 门以上课程的学生学号。

```
SELECT Sno  
FROM SC  
GROUP BY Sno  
HAVING COUNT(*) > 2;
```



例11、查询平均60分以上的学生学号和平均成绩。

```
SELECT Sno, AVG(Grade) FROM SC  
WHERE Grade >= 60  
GROUP BY Sno;
```



逻辑错误

```
SELECT Sno, AVG(Grade) FROM SC  
WHERE AVG(Grade) >= 60  
GROUP BY Sno;
```

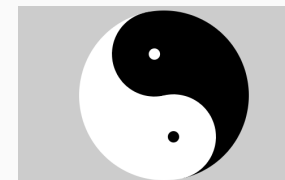


语法错误

```
SELECT Sno, AVG(Grade) FROM SC  
GROUP BY Sno  
HAVING AVG(Grade) >= 60;
```



筛选分组



例12、求‘1’号课程的课程号和课程平均分。

```
SELECT Cno,AVG(Grade)  
FROM SC  
GROUP BY Cno  
HAVING Cno = '1';
```

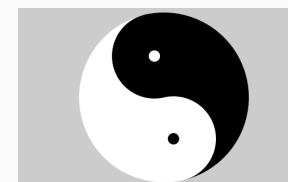


低效率

```
SELECT Cno,AVG(Grade)  
FROM SC  
WHERE Cno = '1'
```



高效率

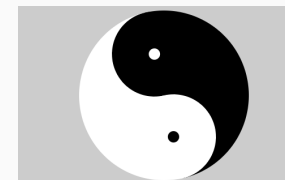


例13、求各门课程的课程号和课程平均分、按分数降序排列。

```
SELECT Cno ,AVG(Grade) FROM SC  
GROUP BY Cno  
ORDER BY AVG(Grade) DESC
```

例14、查询平均85以上的学生学号和平均成绩（60分以上参与计算），并且按照平均成绩排序后取前3条数据显示。

```
SELECT Sno, AVG(Grade) FROM SC WHERE Grade  
> 60  
GROUP BY Sno HAVING AVG(Grade)>=85;  
ORDER BY AVG(Grade)
```



聚合函数：

获取统计信息（总数，求和、平均、最大、最小）

GROUP BY：

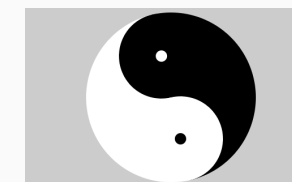
获取分组，配合聚合函数统计分组信息

Having 子句：

筛选分组

提醒：

统计信息查询中不能出现统计内的个体信息查询。



请在`student`表上做练习，表结构如下：

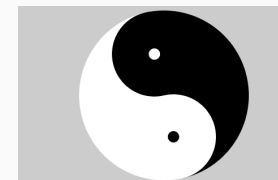
学号 <i>Sno</i>	姓名 <i>Sname</i>	性别 <i>Ssex</i>	年龄 <i>Sage</i>	所在系 <i>Sdept</i>
201215121	李勇	男	20	CS

练习1、查询学生总数。

```
SELECT COUNT(*) AS 学生人数 FROM SC ;
```

练习2、查询最大年龄。

```
SELECT MAX(sage) FROM SC ;
```

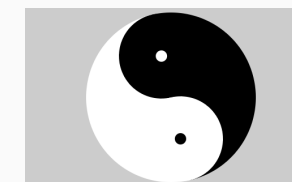


练习3、查询男、女学生的平均年龄。

```
SELECT    Ssex, AVG(sage) AS AG  
FROM      Student  
GROUP BY Ssex;
```

练习4、查询各年龄对应的人数。

```
SELECT    Sage, COUNT(*)  
FROM      Student  
GROUP BY Sage;
```

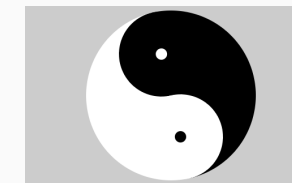


练习5、查询每个系的人数。

```
SELECT Sdept, COUNT(*)  
FROM Student  
GROUP BY Sdept;
```

练习6、查询男学生数多于30的系和系人数，按照人数排序。

```
SELECT Sdept, COUNT(*)  
FROM Student  
WHERE Ssex='男'  
GROUP BY Sdept;  
Having COUNT(*) > 30  
ORDER BY count(*) ;
```



◆ MySQL 和 Navicat的安装使用

◆ 单表查询

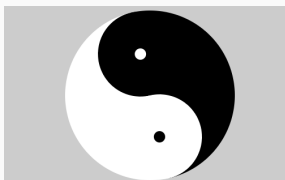
学号 <i>Sno</i>	姓名 <i>Sname</i>	性别 <i>Ssex</i>	年龄 <i>Sage</i>	所在系 <i>Sdept</i>
201215121	李勇	男	20	CS
201215122	刘晨	女	19	CS
201215123	王敏	女	18	MA
200515125	张立	男	19	IS

男女人数
? 平均年
龄?

投影列

筛选行

分组统计



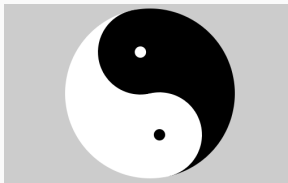
SELECT <目标列表达式> *FROM* <表名>

目标列表达式:

单表选列

- ① 全部列: *
- ② 列名: 列名1, 列名2
- ③ 去掉重复值: *Distinct* 列名
- ④ 列别名: 列名1 [*AS*] 列别名

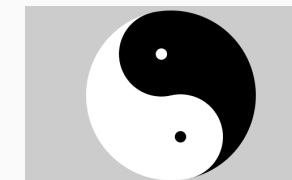
- ⑤ 常量值: ' 中国 ', 100
- ⑥ 表达式: 2022 - 列名
- ⑦ 函数: *YEAR(NOW())*



SELECT 列 FROM 表 WHERE 条件 ;

单表选行

条件	运算符
比较	=, >, <, >=, <=, !=, <>, !>, !<;
组合	AND, OR, NOT
范围	BETWEEN ... AND ..., NOT BETWEEN ... AND ..
集合	... IN(...), ... NOT IN(...)
空值	IS NULL, ... IS NOT NULL
模糊	... LIKE ..., ... NOT LIKE
排序	ORDER BY 列 ASC, ORDER BY 列DESC
分页	LIMIT (SQL server类似关键字为 top)



分组统计

聚合函数:

获取统计信息(总数, 求和、平均、最大、最小)

GROUP BY:

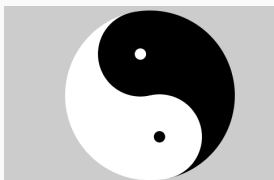
获取分组, 配合聚合函数统计分组信息

Having 子句:

筛选分组

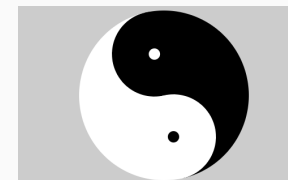
善意提醒:

统计信息查询中不能出现统计内的个体信息查询。



单表查询的一般格式为

```
SELECT [ALL|DISTINCT] <目标列表达式>  
FROM <表名或视图名>  
[WHERE <条件表达式>]  
[GROUP BY <列名> [HAVING <条件表达式>]]  
[ORDER BY <列名> [ASC|DESC]]  
[LIMIT [offset, ] rows]
```



Thank You !